THE COMPANION BY THE NOME: APÉRY'S $\zeta(3)$ NUMBERS AND THE MODULAR ROUTE TO SECOND SOLUTIONS

Abstract. Apéry's proof that $\zeta(3) \notin \mathbb{Q}$ runs on a three-term recurrence with two solutions: the integers a_n, and a companion b_n with $b_n/a_n \to \zeta(3)$. The companion is classically described by a harmonic-sum formula. This paper explains a different route to it — the *modular* route — in which the companion is read off from the nome of the associated differential operator rather than from any sum over binomial cells. We work the Apéry case in full, because there everything is classical and checkable: the mirror map $t(q)$ is Beukers' level-6 eta quotient, the operator factors as $L = (P_3/\sigma^3)\, F\, \theta_q^3\, F^{-1}$, and the second solution becomes a single Eichler-type integral, $b_n = 6\,[t^n]\, F(q)\, \theta_q^{-3}\big(t\sigma^3/(P_3 F)\big)$. All expansions are exact and were verified by machine over $\mathbb{Q}$; the ranges are stated. A closing section explains why this detour is worth learning: in the surrounding programme it produces companions for sporadic families where harmonic-sum formulas provably do not exist.

1. Two solutions and one miracle

In 1978 Apéry announced that

$$\zeta(3) = \sum_{n\geq 1} \frac{1}{n^3}$$

is irrational, and the proof — as relayed by van der Poorten's celebrated report [6] — was so compressed that the audience largely did not believe it. The engine is one recurrence:

$$(n+1)^3 u_{n+1} = (34n^3 + 51n^2 + 27n + 5)\, u_n - n^3 u_{n-1}, \qquad n \geq 1. \tag{1.1}$$

It is convenient to notice that the middle polynomial factors, $34n^3 + 51n^2 + 27n + 5 = (2n+1)(17n^2 + 17n + 5)$; the pair $(17, 5)$ and the trailing n^3 are the only data.

Equation (1.1) has a two-dimensional solution space, and the whole proof is the interplay of two particular members of it.

The first is

$$a_n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2, \qquad a_0 = 1,\ a_1 = 5,\ a_2 = 73,\ a_3 = 1445, \ldots \tag{1.2}$$

That these integers satisfy (1.1) is the first surprise (they are a sum over $n+1$ cells, and the recurrence knows nothing about cells); it is nowadays a two-line consequence of Zeilberger's algorithm, and we take it as known. [verified: $n \leq 25$, exactly]

The second solution is the companion b_n, pinned down by

$$b_n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2 \left(\sum_{m=1}^{n} \frac{1}{m^3} + \sum_{m=1}^{k} \frac{(-1)^{m-1}}{2m^3 \binom{n}{m}\binom{n+m}{m}} \right), \tag{1.3}$$

so that $b_0 = 0$ and $b_1 = 6$. We use this normalisation throughout; it is the one in [6]. [verified: (1.3) satisfies (1.1) for $1 \leq n \leq 24$, exactly]

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n	a_n	b_n
0	1	0
1	5	6
2	73	351/4
3	1445	62531/36
4	33001	11424695/288
5	819005	35441662103/36000
6	21460825	20637706271/800

n	b_n/a_n
1	1.2
2	1.20205479452
3	1.20205690119
4	1.20205690316
5	1.20205690316
6	1.20205690316
∞	$\zeta(3) = 1.20205690316$

TABLE 1. The two solutions, and the convergence. Both columns are exact rationals; the quotients are shown to 12 digits. [verified: computed]

The miracle is that these two sequences approach each other's quotient extremely fast. The two roots of $t^2 - 34t + 1$ are $\alpha^{\pm 1}$ with

$$\alpha = (1+\sqrt{2})^4 = 17 + 12\sqrt{2} = 33.97\ldots, \tag{1.4}$$

these are the reciprocal singularities of (1.1), and the classical asymptotics (see [6, 3]) are

$$a_n \sim C\,\alpha^n n^{-3/2}, \qquad \bigl|b_n - a_n\zeta(3)\bigr| = O\bigl(\alpha^{-n}\bigr), \qquad \text{so} \qquad \lim_{n\to\infty} \frac{b_n}{a_n} = \zeta(3). \tag{1.5}$$

Let $d_n = \operatorname{lcm}(1,\ldots,n)$, so $d_n = e^{n(1+o(1))}$ by the prime number theorem. Three inputs finish the proof.

(i) $a_n \in \mathbb{Z}$ (clear from (1.2)) and $d_n^3 b_n \in \mathbb{Z}$. The second is *not* clear from (1.3) — the inner sum there carries binomial coefficients in its denominators — and is a genuine arithmetic theorem; see [6, §3] and [3].

(ii) $b_n - a_n\zeta(3) \neq 0$ for all large n. This too needs an argument; the cleanest is Beukers' integral representation [3], in which $b_n - a_n\zeta(3)$ is (up to sign) a triple integral of a strictly positive integrand, hence nonzero.

(iii) $e^3 < \alpha$, i.e. $20.08\ldots < 33.97\ldots$

Now suppose $\zeta(3) = p/q$ with $p, q \in \mathbb{Z}$, $q \geq 1$. Then $q\,d_n^3\bigl(b_n - a_n\zeta(3)\bigr)$ is an integer by (i); it is nonzero by (ii); and by (1.5) and (iii) its absolute value is $O\bigl(q\,(e^3/\alpha)^{n(1+o(1))}\bigr) \to 0$. A nonzero integer cannot tend to 0. Hence $\zeta(3) \notin \mathbb{Q}$; that is the whole proof.

Table 1 is the whole story in numbers, and the reader who has never seen it should stop and admire the right-hand column: six terms of a recurrence give ten digits of $\zeta(3)$.

The question this paper answers. Formula (1.3) is a *harmonic-sum* formula: same binomial cells as a_n, weighted by harmonic-type numbers. It is a beautiful thing, and for a long time it was the only known way to write a companion down. But it is a formula of a very particular shape, and that shape is not always available: in the surrounding research programme (§7) there are Apéry-like families for which one can *prove* that no formula of type (1.3) exists, and yet the companion is a perfectly concrete sequence with a perfectly concrete arithmetic. We want a construction of the second solution that does not go through cells at all.

There is one, and it is modular. The route is:

recurrence $\rightsquigarrow$ differential operator $\rightsquigarrow$ Frobenius basis $\rightsquigarrow$ nome q $\rightsquigarrow$ modular parametrisation $\rightsquigarrow$ companion.

Every arrow is classical for Apéry; only the last one is recent, and it is the one we will derive in §6. The Apéry case is our worked example precisely because the answer is already known: it lets us check every step.

2. From recurrence to operator

Set $\theta_t = t\frac{d}{dt}$, so that $\theta_t t^n = nt^n$. Any three-term recurrence whose coefficients are polynomial in n becomes a differential operator by the substitution $n \mapsto \theta_t$, shift $\mapsto$ multiplication by t. For (1.1), with $(a, b, c, d) = (17, 5, 1, 0)$:

Definition 2.1.

$$L \;=\; \theta_t^3 \;-\; t\,(2\theta_t + 1)(a\theta_t^2 + a\theta_t + b) \;+\; t^2(\theta_t + 1)\big(c(\theta_t + 1)^2 + d\big).$$

Expanding L on a formal series $y = \sum_n u_n t^n$ and reading the coefficient of t^n gives exactly

$$[t^n]\,L(y) = n^3 u_n - (2n-1)\big(a(n-1)^2 + a(n-1) + b\big)u_{n-1} + (n-1)^3 u_{n-2},$$

which is (1.1) with n replaced by $n-1$. So:

Observation 2.2. $L(y) = 0$ if and only if the coefficients of y satisfy (1.1) *for every* $n \geq 0$, including $n = 0$. The $n = 0$ instance reads $1^3 u_1 = 5u_0$, and it is the one piece of bookkeeping that will later earn its keep: see §6.

Thus $y_0(t) := \sum_{n\geq 0} a_n t^n$ satisfies $L(y_0) = 0$ ([verified: to t^{26}, exactly]), while $y_b := \sum_n b_n t^n$ does *not*: b satisfies (1.1) only for $n \geq 1$, and the $n = 0$ defect is $1^3 b_1 - 5b_0 = 6$. Hence

$$L(y_b) = 6t. \tag{2.1}$$

This is not a blemish. It is the entire mechanism by which the second solution is *selected*: y_0 is the kernel, y_b is the solution of an inhomogeneous problem whose right-hand side is one monomial.

The indicial polynomial of L at $t = 0$ is the θ_t-part with $t = 0$, namely θ_t^3: a triple root at 0. In the language of Picard–Fuchs equations, L has *maximal unipotent monodromy* (MUM) at the origin. This is the structural fact everything below rests on. It says the local solution space is spanned by

$$y_0, \qquad y_1 = y_0 \log t + g, \qquad y_2 = \tfrac{1}{2} y_0 \log^2 t + g \log t + h, \tag{2.2}$$

with $g, h \in t\mathbb{Q}[[t]]$ — one holomorphic solution and two logarithmic ones, with no fractional exponents anywhere.

Computing g is elementary. Write $L = \sum_{j=0}^{2} t^j P_j(\theta_t)$ as in Definition 2.1. Because $\theta_t(f \log t) = (\theta_t f)\log t + f$ we get $P(\theta_t)(f \log t) = (P(\theta_t)f)\log t + P'(\theta_t)f$, and t^j commutes with all of this, so (2.2) forces

$$L(g) = -\sum_{j=0}^{2} t^j P_j'(\theta_t)\, y_0. \tag{2.3}$$

Since $P_0(\theta_t) = \theta_t^3$ is invertible on t^n for $n \geq 1$, (2.3) determines g coefficientwise with no free constant beyond $g_0 = 0$. One turn of the crank gives [verified: exact]

$$g(t) = 12t + 210t^2 + 4438t^3 + 104825t^4 + \tfrac{13276637}{5}t^5 + 70543291\,t^6 + \tfrac{67890874657}{35}t^7 + \cdots \tag{2.4}$$

Remark 2.3. g is not b and is not close to it: $g_1 = 12$ while $b_1 = 6$. The relation between g and the companion is exactly the boundary bookkeeping of (2.1), and getting it right is the content of §6. It is worth flagging early because the naive guess "$b_n = g_n + \lambda a_n$" is false, and was tried.

3. The nome, and why one would invent it

Under MUM there is a canonical coordinate. Consider the quotient y_1/y_0; by (2.2) it is $\log t + g/y_0$, a well-defined element of $\log t + t\mathbb{Q}[[t]]$. Exponentiate:

Definition 3.1 (the nome)**.**

$$q(t) \;:=\; \exp\bigl(y_1/y_0\bigr) \;=\; t\exp\bigl(g(t)/y_0(t)\bigr) \;\in\; t+t^2\mathbb{Q}[[t]].$$

Its compositional inverse $t(q)\in q+q^2\mathbb{Q}[[t]]$ is the *mirror map*.

Why is this the right thing to do? Two answers, one formal and one geometric.

Formally: the monodromy of L around $t=0$ is $y_1\mapsto y_1+2\pi i\,y_0$, i.e. $y_1/y_0\mapsto y_1/y_0+2\pi i$, i.e. $q\mapsto q$. The nome is the invariant coordinate — the coordinate in which the local monodromy has been divided out. Whatever transcendental object is hiding behind L, it is a function of q, not of t.

Geometrically: for Picard–Fuchs equations of families of Calabi–Yau type, y_1/y_0 is a period ratio, so $q=e^{2\pi i\tau}$ with τ the point in the upper half plane classifying the fibre. Then $t(q)$ is a modular function and $y_0(t(q))$ a modular form — *if* the family is of the sort where τ actually lives on a modular curve. For L of order 3 this is the symmetric-square situation and it happens genuinely often; Apéry is the prototype.

Cranking out Definition 3.1 from (2.4) [verified: exact, to q^{26}]:

$$q(t)=t+12t^2+222t^3+4900t^4+119133t^5+3079152t^6+\cdots, \tag{3.1}$$

$$t(q)=q-12q^2+66q^3-220q^4+495q^5-804q^6+1068q^7+\cdots \tag{3.2}$$

Two things should now make the reader sit up. First, $t(q)$ has *integer* coefficients — there is no reason on earth for that in Definition 3.1, which involves dividing by y_0, exponentiating and reverting a series, all operations that manufacture denominators freely (and indeed g itself has a 5 and a 35 in its denominators, (2.4)). Integrality of the mirror map is the fingerprint of modularity — with the caveat, which we will not need but which should be stated once and loudly, that integral mirror maps are known well outside the modular-curve setting (the quintic threefold's is integral, and no modular curve is in sight). Integrality is *evidence that prompts an identification*; the identification is what proves something, and it is Theorem 4.1 that does the work below. Second, the initial coefficients $1,-12,66,-220,495$ are $\binom{12}{0},-\binom{12}{1},\binom{12}{2},-\binom{12}{3},\binom{12}{4}$: the series is trying to tell us it is a twelfth power of something starting $1-q$.

4. Beukers' parametrisation

It is. With the Dedekind eta function $\eta(q)=q^{1/24}\prod_{n\ge1}(1-q^n)$ (we write q rather than τ in the argument, $q=e^{2\pi i\tau}$), define

$$\mathbf{t}(q)=\left(\frac{\eta(q)\,\eta(q^6)}{\eta(q^2)\,\eta(q^3)}\right)^{12}, \qquad \mathbf{F}(q)=\frac{\bigl(\eta(q^2)\,\eta(q^3)\bigr)^7}{\bigl(\eta(q)\,\eta(q^6)\bigr)^5}. \tag{4.1}$$

The fractional powers of q in the eta prefactors work out to $q^{12(1+6-2-3)/24}=q^1$ for $\mathbf{t}$ and $q^{(5\cdot7-7\cdot5)/24}=q^0$ for $\mathbf{F}$, so $\mathbf{t}\in q+q^2\mathbb{Z}[[q]]$ and $\mathbf{F}\in 1+q\mathbb{Z}[[q]]$, as they must be. (This prefactor bookkeeping is worth doing once by hand: a stray q in front of $\mathbf{t}$ is the most common way to misquote (4.1).)

Theorem 4.1 (Beukers [2])**.** *$t(q)=\mathbf{t}(q)$ and $F(q):=y_0(t(q))=\mathbf{F}(q)$. Equivalently: t is a hauptmodul for the group $\Gamma_0(6)+6$ (i.e. $\Gamma_0(6)$ extended by the Fricke involution $\tau\mapsto-1/6\tau$), the mirror map is the corresponding uniformisation, and y_0 pulls back to a weight-2 modular form on $\Gamma_0(6)$.*

We do not reprove this — it is Beukers' theorem, and the modern statement of the surrounding picture is Zagier's [7] together with [1, 5] — but we do check it, because checking it is ten lines of code and it is the identity everything else rests on.

Proposition 4.2 (machine check)**.** *With $t(q)$, $F(q)$ computed from Definition 3.1 out of the recurrence* (1.1) *alone, and $\mathbf{t}, \mathbf{F}$ expanded from* (4.1) *by the product formula, one has $t(q) = \mathbf{t}(q)$ and $F(q) = \mathbf{F}(q)$ coefficientwise over $\mathbb{Q}$.* [verified: to q^{26}, exact]

The first coefficients, for the reader's own ten lines:

$$t(q) = q - 12q^2 + 66q^3 - 220q^4 + 495q^5 - 804q^6 + \cdots, \tag{4.2}$$

$$F(q) = 1 + 5q + 13q^2 + 23q^3 + 29q^4 + 30q^5 + 31q^6 + 40q^7 + \cdots \tag{4.3}$$

Note F's coefficients are small — a weight-2 form has coefficients $O(n^{1+\epsilon})$ — whereas a_n grows like α^n. All of the growth of the Apéry numbers has been absorbed into the change of variable. That is the practical reason this coordinate is worth the trouble: in q, the problem is bounded.

Remark 4.3 (the Apéry numbers, recovered)*.* Reading Theorem 4.1 backwards, $\sum a_n t^n = \mathbf{F}(q)$ where q is the inverse of $\mathbf{t}$: the Apéry numbers are the Taylor coefficients of a weight-2 eta quotient in the coordinate given by a weight-0 one. Their integrality — obvious from (1.2), mysterious from (1.1) — becomes, from this side, the statement that both eta quotients in (4.1) are integral q-series. Every "Apéry-like" integrality phenomenon in the literature has, conjecturally or provably, this shape.

5. The operator, factored through the nome

Now we exploit the coordinate. Write $\theta_q = q\frac{d}{dq}$ and

$$\sigma \ := \ \theta_q \log t \ = \ \frac{q\,t'(q)}{t(q)} \ \in \ 1 + q\mathbb{Z}[[q]], \qquad \text{so} \qquad \theta_q = \sigma\,\theta_t \tag{5.1}$$

as operators on functions of t (chain rule). Concretely [verified: exact, q^{26}]

$$\sigma = 1 - 12q - 12q^2 - 12q^3 - 12q^4 - 72q^5 - 12q^6 - 96q^7 - \cdots \tag{5.2}$$

and, pleasantly, taking $\theta_q \log$ of (4.1) and using $\theta_q \log \eta(q^m) = \frac{m}{24}E_2(q^m)$, where $E_2 = 1 - 24\sum_{n\geq 1}\bigl(\sum_{d|n} d\bigr)q^n$ is the quasimodular Eisenstein series of weight 2, gives the closed form

$$\sigma = \tfrac{1}{2}\bigl(E_2(q) + 6E_2(q^6) - 2E_2(q^2) - 3E_2(q^3)\bigr), \tag{5.3}$$

a weight-2 Eisenstein series on $\Gamma_0(6)$. [verified: to q^{25}, exact] So σ is modular data too, of the same weight as F.

Let $P_3(t)$ be the coefficient of θ_t^3 in L, i.e. the leading symbol: from Definition 2.1,

$$P_3(t) = 1 - 2at + ct^2 = 1 - 34t + t^2, \tag{5.4}$$

whose roots $17 \pm 12\sqrt{2}$ are (inverses of) the singular points of L, and whose larger root is precisely the α governing the convergence rate in (1.4). Everything in Apéry's proof is in this quadratic.

An interlude that cannot be skipped: the symmetric square. It is tempting — and it is wrong — to think that the factorisation below follows from MUM alone. Let us be exact about where the extra hypothesis enters.

MUM gives the Frobenius shape (2.2), in which h is uniquely determined; it does *not* say what h is. The factorisation we are after requires the specific value

$$h = \frac{g^2}{2y_0}, \qquad \text{equivalently} \qquad y_2 = \frac{y_1^2}{2y_0}, \qquad \text{equivalently} \qquad 2y_0 h - g^2 = 0. \tag{5.5}$$

For a general order-3 MUM operator (5.5) is false. The condition that makes it true is that L be a *symmetric square*: if u, v span the solution space of a second-order operator D, then u^2, uv, v^2 span a three-dimensional space, and the order-3 operator annihilating it is $\mathrm{Sym}^2(D)$. On such an operator, $y_0 = u^2$, $y_1 = uv$, $y_2 = v^2/2$, and (5.5) is the identity $2 \cdot u^2 \cdot \frac{v^2}{2} = (uv)^2$ — trivially true, and a real restriction.

Apéry's operator does satisfy it, and we prove it rather than cite it. Normalise L to be monic in $\partial = d/dt$:

$$\frac{1}{t^3 P_3(t)}\, L = \partial^3 + A_2\partial^2 + A_1\partial + A_0.$$

The classical formula for the symmetric square of $D = \partial^2 + p\partial + r$ is

$$\mathrm{Sym}^2(D) = \partial^3 + 3p\,\partial^2 + (2p^2 + p' + 4r)\,\partial + (4pr + 2r'). \tag{5.6}$$

So L is a symmetric square iff, setting $p := A_2/3$ and $r := \frac{1}{4}(A_1 - 2p^2 - p')$ — forced by matching the ∂^2 and ∂ coefficients — the remaining equation $A_0 = 4pr + 2r'$ holds identically in $\mathbb{Q}(t)$.

Lemma 5.1 (L is a symmetric square)**.** *With $P_3 = 1 - 34t + t^2$, put*

$$p = \frac{2t^2 - 51t + 1}{t\,P_3(t)}, \qquad r = \frac{t - 10}{4t\,P_3(t)}, \qquad D = \partial^2 + p\,\partial + r.$$

Then $L = t^3 P_3(t) \cdot \mathrm{Sym}^2(D)$, exactly, as operators over $\mathbb{Q}(t)$. Equivalently, in θ_t-form,

$$\widetilde{D} \;=\; \theta_t^2 - t\big(34\theta_t^2 + 17\theta_t + \tfrac{5}{2}\big) + t^2\big(\theta_t + \tfrac{1}{2}\big)^2, \tag{5.7}$$

which is again MUM (indicial polynomial θ_t^2).

Proof. Expand L in ∂: from Definition 2.1,

$$L = t^3(t^2 - 34t + 1)\partial^3 + 3t^2(2t^2 - 51t + 1)\partial^2 + t(7t^2 - 112t + 1)\partial + t(t - 5).$$

Divide by $t^3 P_3$, read off $p = A_2/3$ and $r = \frac{1}{4}(A_1 - 2p^2 - p')$ — these are the values in the statement — and verify $A_0 - (4pr + 2r') = 0$ by clearing denominators. This is a finite rational-function identity; it was checked symbolically over $\mathbb{Q}(t)$ and the residual is identically 0. [verified: exact, symbolic over $\mathbb{Q}(t)$; check (C12a)] ☐

Corollary 5.2. *$y_2 = y_1^2/(2y_0)$, i.e.* (5.5) *holds.*

Proof. $\widetilde{D}$ is MUM of order 2, so its local solutions at 0 are $u \in 1 + t\mathbb{Q}[[t]]$ and $v = u\log t + w$ with $w \in t\mathbb{Q}[[t]]$, both unique. By Lemma 5.1, u^2 is a holomorphic solution of L with value 1 at $t = 0$; under MUM the holomorphic solution space of L is one-dimensional, so $u^2 = y_0$. Then $uv = u^2\log t + uw = y_0\log t + uw$ is a solution of L of the Frobenius shape (2.2), so by uniqueness $y_1 = uv$ and $g = uw$; and $v^2/2 = \frac{1}{2}y_0\log^2 t + g\log t + \frac{1}{2}w^2$ likewise gives $y_2 = v^2/2$ and $h = w^2/2$. Hence $2y_0h - g^2 = u^2w^2 - (uw)^2 = 0$. ☐

For the reader who wants to see them, $u = 1 + \frac{5}{2}t + \frac{267}{8}t^2 + \frac{10225}{16}t^3 + \cdots$, and $u^2 = y_0$, $uw = g$ hold coefficientwise. [verified: to t^{26}, exact; checks (C12b–c)]

The factorisation.

Proposition 5.3 (the factorisation)**.** *Let L be an order-3 MUM operator satisfying the symmetric-square condition* (5.5) *— for Apéry's L this is Corollary 5.2. Then, as operators on functions of t near $t = 0$,*

$$L \;=\; \frac{P_3(t)}{\sigma^3} \;\cdot\; F \;\cdot\; \theta_q^3 \;\cdot\; F^{-1}, \tag{5.8}$$

where P_3 is the leading symbol of L in θ_t, and $F = F(q)$, σ, q are as above; $F^{\pm 1}$ denote multiplication operators.

Proof. Both sides are linear differential operators of order 3 in t; we show they have the same kernel and the same leading symbol.

Kernel. θ_q^3 kills exactly $1, \log q, \frac{1}{2}\log^2 q$. Hence $M := F\theta_q^3 F^{-1}$ kills exactly F, $F\log q$, $\frac{1}{2}F\log^2 q$. Now $\log q = y_1/y_0$ by Definition 3.1, and $F = y_0$ as functions (Theorem 4.1 is not needed here: F is *defined* as $y_0 \circ t$). So

$$F = y_0, \quad F\log q = y_0 \cdot \frac{y_1}{y_0} = y_1, \quad \tfrac{1}{2}F\log^2 q = \frac{y_1^2}{2y_0} = y_2,$$

the last step being exactly the symmetric-square condition (5.5). So $\ker M = \ker L$, both being the span of y_0, y_1, y_2. (Equivalently, and more transparently: with u, v as in Corollary 5.2, $\log q = v/u$ and $F = u^2$, so $F \cdot \{1, \log q, \frac{1}{2}\log^2 q\} = \{u^2, uv, \frac{1}{2}v^2\}$, which is $\ker \mathrm{Sym}^2(D) = \ker L$.)

Leading symbol. Two order-3 operators with the same kernel differ by a left multiplier, so $L = \lambda(t)\, M$ for some function λ. By (5.1), $\theta_q^3 = (\sigma\theta_t)^3 = \sigma^3\theta_t^3 + (\text{order} \le 2)$, and conjugating by the multiplication operator F does not change the leading symbol. Hence the θ_t^3-coefficient of M is σ^3, while that of L is $P_3(t)$. Therefore $\lambda = P_3/\sigma^3$. □

The content of (5.8) is worth saying in words. *In the nome coordinate, and after dividing out the modular form F, Apéry's operator is just θ_q^3.* All the arithmetic — the $34n^3 + 51n^2 + 27n + 5$, the $\sqrt{2}$, the growth rate — has been pushed into the change of variable $t \leftrightarrow q$ and into the prefactor. What remains is the simplest operator of order 3 there is.

Remark 5.4 (the Yukawa coupling, and what happens without (5.5)). It is worth knowing what (5.8) looks like when the symmetric-square hypothesis fails, because the failure is not catastrophic — it is *measured*. For a general order-3 MUM operator, passing to the canonical coordinate still normalises the operator, but the normal form is $F\,\theta_q\,K^{-1}\,\theta_q\,\theta_q\,F^{-1}$ (up to a left multiplier), where the extra function K is the *Yukawa coupling* — in mirror-symmetry language, the generating series of instanton numbers, and in ours simply the obstruction $2y_0h - g^2$ repackaged. Condition (5.5) is exactly K constant, in which case K can be absorbed and the bare θ_q^3 survives. So Theorem 6.1 below is a statement about the symmetric-square class, and the general statement carries one more datum.

This is not idle. The programme surrounding this paper finds precisely that dichotomy empirically: all fifteen sporadic Apéry-like families of [7, 1, 4] satisfy the bare formula, while the order-5 $\zeta(5)$ block studied in [9] does not — it is Sym^4-like rather than Sym^2-like, bare θ_t-powers are excluded there, and it carries a nontrivial integral Yukawa-type invariant. I thank the referee for insisting on this distinction, which the first draft elided.

And θ_q^3 can be inverted on $q\mathbb{Q}[[q]]$ by inspection:

$$\theta_q^{-3}\Big(\sum_{m\ge1}\Psi_m q^m\Big) = \sum_{m\ge1}\frac{\Psi_m}{m^3}q^m. \tag{5.9}$$

The reader will recognise the right-hand side as an *Eichler integral*: a formal q-series obtained from a weight-k object by dividing the m-th coefficient by m^{k-1}. Here $k-1=3$. We come back to this in §7; for now, note that $\zeta(3) = \sum m^{-3}$ has just walked into the room by itself.

6. The companion, derived

We can now do the thing the harmonic formula (1.3) does, without cells.

Recall Observation 2.2 and (2.1). It is cleaner to use the *unit-normalised* second solution

$$B(0) = 0, \qquad B(1) = 1, \qquad B \text{ satisfies (1.1) } (n \ge 1),$$

so that $b_n = 6B(n)$ [verified: $n \le 25$, exact] and, by the same one-line defect computation, $y_B := \sum_n B(n)t^n$ satisfies

$$L(y_B) = t. \tag{6.1}$$

[verified: $L(y_B) = t$ exactly, all coefficients to t^{26}]

Theorem 6.1 (the companion by the nome)**.** *Let L be an order-3 MUM operator satisfying the symmetric-square condition (5.5), so that Proposition 5.3 applies; for Apéry's L this is Lemma 5.1 and Corollary 5.2. Let $F, \sigma, P_3, t(q)$ be as above and set*

$$\Psi(q) \;:=\; \frac{t(q)\,\sigma(q)^3}{P_3\big(t(q)\big)\,F(q)} \;\in\; q\mathbb{Q}[[q]], \qquad \Theta(q) := \theta_q^{-3}\Psi = \sum_{m\ge 1} \frac{\Psi_m}{m^3}\, q^m. \tag{6.2}$$

Then the second solution of (1.1) is

$$B(n) = [t^n]\; F(q)\,\Theta(q) \qquad \big(\textit{expanded in } t \textit{ via } q = q(t)\big), \qquad b_n = 6B(n). \tag{6.3}$$

Proof. Substitute (5.8) into (6.1): $\frac{P_3}{\sigma^3} F\,\theta_q^3(y_B/F) = t$, i.e. $\theta_q^3(y_B/F) = t\sigma^3/(P_3F) = \Psi$. The right-hand side lies in $q\mathbb{Q}[[q]]$ (it has a simple zero at $q = 0$ since $t = q + O(q^2)$ and $\sigma, P_3^{-1}, F^{-1} \in 1 + q\mathbb{Q}[[q]]$), so θ_q^{-3} applies by (5.9) and returns the unique solution in $q\mathbb{Q}[[q]]$; the three constants of integration are exactly the kernel $\{F, F\log q, \frac{1}{2}F\log^2 q\}$ of θ_q^3 after multiplying back by F, and they are fixed by $y_B \in t\mathbb{Q}[[t]]$ (no logarithms, no constant term). Hence $y_B = F\cdot\theta_q^{-3}\Psi$. □

That is the whole construction. It is worth restating what has been used: the recurrence (to build L), MUM (to get the Frobenius basis and hence q), the factorisation (Proposition 5.3), and the boundary defect (6.1) — *the $n = 0$ instance of the recurrence, which the second solution is precisely the one to violate.* Nothing about binomial coefficients; nothing about harmonic numbers. If the family is modular, this runs.

The numbers. Here is the computation in full, so that a reader can reproduce it. From (4.2), (4.3), (5.2) and $P_3 = 1 - 34t + t^2$:

$$\Psi(q) = q - 19q^2 + 91q^3 - 179q^4 + 126q^5 - q^6 + 344q^7 - 1459q^8 + \cdots, \tag{6.4}$$

$$\Theta(q) = q - \tfrac{19}{8}q^2 + \tfrac{91}{27}q^3 - \tfrac{179}{64}q^4 + \tfrac{126}{125}q^5 - \tfrac{1}{216}q^6 + \cdots \tag{6.5}$$

(the second line is the first divided by m^3, term by term — that is all θ_q^{-3} is). Multiplying by F and re-expanding in t via (3.1):

n	$[t^n]F\Theta$	$B(n)$ from (1.1)	$b_n = 6B(n)$
0	0	0	0
1	1	1	6
2	117/8	117/8	351/4
3	62531/216	62531/216	62531/36
4	11424695/1728	11424695/1728	11424695/288
5	35441662103/216000	35441662103/216000	35441662103/36000
6	20637706271/4800	20637706271/4800	20637706271/800

Table 2. Theorem 6.1 against the recurrence. The two middle columns agree for all $n \le 25$, exactly over $\mathbb{Q}$. [verified: $n \le 25$]

The last column of Table 2 is the last column of Table 1: the modular route reproduces van der Poorten's b_n on the nose. [verified: $n \le 25$, exact]

Remark 6.2 (reading the answer). The three ingredients of Ψ each have a modular meaning: t is the hauptmodul, σ is the weight-2 Eisenstein series (5.3), F is the weight-2 form of Theorem 4.1, and $P_3(t) = 1 - 34t + t^2$ is a polynomial in the hauptmodul, i.e. again a modular function. Counting weights (3×2 from σ^3, -2 from F^{-1}, 0 from t and P_3), Ψ transforms with weight 4 on $\Gamma_0(6)$.

Two honest caveats. First, σ is quasimodular, not modular ((5.3) is a combination of E_2's, and only the particular combination there — whose E_2-anomalies cancel — is genuinely weight 2); we have checked that cancellation only through (5.3)'s coefficient identity. Second, and more to the point, "weight 4" here means *meromorphic* of weight 4: dividing by F and by $P_3(t)$ can and does introduce poles wherever those vanish, and establishing holomorphy would require checking the zeros of F and the zeros of $P_3 \circ t$ against the cusps of $\Gamma_0(6)$, which we have not done. Everything in Theorem 6.1 is a statement about formal q-series near $q = 0$, where all the inverted quantities are units, and it is valid as such unconditionally. The global modular reading is a (correct-looking, unverified here) interpretation on top.

With that said, Theorem 6.1 reads: *the companion of the Apéry numbers is F times the Eichler integral of an explicit weight-4 object.* The little coefficients of (6.4) are the letters this alphabet writes with. (Amusing: $\Psi_6 = -1$.)

Remark 6.3 (why the harmonic formula and this one are both right). Two formulas for the same sequence must be an identity. Comparing (1.3) and (6.3) coefficientwise gives, for each n, a nontrivial identity between a cell sum with harmonic weights and a q-expansion coefficient of a modular Eichler integral. The reader is invited to write out $n = 2$ (351/4 both ways) by hand; it is a satisfying half page, and it explains why nobody guessed one formula from the other.

7. Why one would want the second route

If the harmonic formula (1.3) already exists, the modular derivation is a curiosity for Apéry. The reason to learn it is that it survives where (1.3) does not.

Apéry's recurrence is one member of a now-classical zoo. Zagier's search [7] over the two-parameter family of order-2 analogues turned up exactly six "sporadic" integral solutions; Almkvist–Zudilin [1] and Cooper [4] extended the census at the $\zeta(3)$ level to fifteen sporadic pairs in total. Every one of them has a second solution, defined by the same normalisation $B(0) = 0$, $B(1) = 1$, and every one of them has an Apéry-type limit or an arithmetic reason not to. The natural first question is: what is $B(n)$?

For most of the fifteen, a formula of harmonic type — same cells, weighted by H-sums — can be found, and the working papers of the surrounding programme [11] give the current census of which. But for a definite subfamily the harmonic ansatz can be *closed out*: one writes down the complete tame ansatz — all products of harmonic sums of the right weight over the given cells — and proves, by exact linear algebra over enough n, that the solution space is empty. For those families no formula of shape (1.3) exists at all, and this is a theorem about the ansatz, not a failure of ingenuity.

Precisely there, the construction of §6 works. The programme's modular-anchor probe [8] runs the pipeline of §§2–5 on each unreachable family: it finds that the mirror map $t(q)$ is again integral (so the family is modular), it identifies $F(q)$ where it can — for one such family $F(q) = \big(9E_2(q^9) - E_2(q)\big)/8$, a weight-2 Eisenstein series on $\Gamma_0(9)$; for another, $F(q) = \eta(-q)^3/\eta(-q^3)$, a sign-twisted Borwein cubic theta — and then applies Theorem 6.1 verbatim. Out comes a companion formula in modular letters for a family where the harmonic letters provably do not suffice. The Apéry case of this paper is that programme's control experiment: same code, known answer, Table 2. Related anchors and the variational route to the same data are in [10].

The structural moral is worth a paragraph on its own. Why should a second solution of a recurrence be an *Eichler integral*? Because that is what θ_q^{-3} is, and θ_q^{-3} is what a MUM operator becomes in its own canonical coordinate. Eichler integrals are the classical home for objects that are "one integration away from modular": $\sum a_m m^{1-k} q^m$ for $f = \sum a_m q^m$ of weight k; they are not modular themselves, but their failure to be modular is a period polynomial, and the periods are exactly the special values one is chasing. In our case $\zeta(3) = \sum m^{-3}$ appears as the value the Eichler integral of the trivial (Eisenstein) part contributes at the relevant cusp, and the Apéry limit (1.5) should be the statement that the $q \to$ cusp asymptotics of $F\Theta$ against F is that period. Golyshev–Zagier's "gamma conjecture" framework [5] and the theory of periods of the associated motive are the correct general setting; the elementary version, which is all this paper needs, is Theorem 6.1.

Let us be scrupulous about what has and has not been shown, since the previous paragraph is the one place where enthusiasm could outrun the argument. *This paper constructs the companion modularly; it does not derive the limit $b_n/a_n \to \zeta(3)$ from modular transformation theory.* Doing that would mean evaluating the Eichler integral Θ at the relevant cusp and identifying the resulting period with $\zeta(3)$ — a genuine computation in the theory of periods of Eichler integrals, and one we have not performed. The limit (1.5) is used here exactly as Apéry used it: as a classical fact about the recurrence. The modular story explains the *shape* of the companion; the $\zeta(3)$ itself remains, in this account, an input.

The practical upshot for a working reader: if you have an integral Apéry-like recurrence and want the second solution, do not start by guessing harmonic weights. Compute g, compute q, look at whether $t(q)$ is integral. If it is, you are ten lines of code from the companion, and those ten lines are in the appendix.

8. The checks

Nothing in this paper was asserted without computing it. All arithmetic is exact over $\mathbb{Q}$ (Python `Fractions`); series are truncated at q^{26}, t^{26}. The script is `verify.py` alongside this file; its output is:

- (C1) a_n from (1.2) equals a_n from (1.1), $n \le 25$. PASS
- (C2) b_n from (1.3) satisfies (1.1) for $1 \le n \le 24$, and $b_0 = 0$, $b_1 = 6$; the $n = 0$ defect is $1^3 b_1 - 5b_0 = 6$. PASS
- (C3) b_{25}/a_{25} agrees with $\zeta(3)$ to 40 decimal digits. PASS
- (C4) $t(q) = \big(\eta_1\eta_6/\eta_2\eta_3\big)^{12}$ to q^{26}. PASS
- (C5) $F(q) = (\eta_2\eta_3)^7/(\eta_1\eta_6)^5$ to q^{26}. PASS
- (C6) $B(n) = [t^n]F\,\theta_q^{-3}\Psi$ equals the recurrence's B, $n \le 26$; stated in the text as $n \le 25$ because q^{26} is the truncation edge of the σ construction. PASS
- (C7) $b_n = 6B(n)$, $n \le 25$. PASS
- (C8) $L(y_B) = t$ exactly: coefficient 1 at t^1, zero at every other t^n, $n \le 26$. PASS
- (C9) $L(y_0) = 0$ to t^{26}. PASS
- (C10) the eta prefactor bookkeeping of (4.1) (exponents q^1 and q^0). PASS
- (C11) $\sigma = \frac{1}{2}(E_2(q) + 6E_2(q^6) - 2E_2(q^2) - 3E_2(q^3))$ to q^{25}. PASS
- (C12a) the symmetric-square residual $A_0 - (4pr + 2r')$ of Lemma 5.1 is identically 0 in $\mathbb{Q}(t)$ — symbolic, not truncated. PASS
- (C12b) $u^2 = y_0$ to t^{26}, where u is the holomorphic solution of $\widetilde{D}$. PASS
- (C12c) $u\,w = g$ to t^{26}, where w is $\widetilde{D}$'s log-tail; hence $h = w^2/2$ and $2y_0h - g^2 = 0$. PASS

Everything labelled [verified: …] in the text refers to this list. Theorem 4.1 is quoted from the literature; (C4)–(C5) are a check of it, not a proof. Propositions 5.3 and Theorem 6.1 are proved in the text, and (C6)–(C8) are their independent numerical confirmation. (C12a) is the only check that is a proof rather than evidence: it is an identity between rational functions, verified symbolically, and Lemma 5.1 rests on it. (C12b–c) are consistency checks on Corollary 5.2, which is itself proved.

Appendix A. Ten lines of code

The following is the skeleton; `smul`, `sinv`, `sexp`, `srevert`, `compose` are the obvious truncated power-series operations on lists of `Fractions` (product, reciprocal, exponential, compositional inverse, composition), each five lines.

```
from fractions import Fraction as F
N = 26; a,b,c,d = 17,5,1,0

# 1. the Apery numbers, from the recurrence
A = [F(1), F(5)]
for n in range(1, N+2):
    A.append((F((2*n+1)*(a*n*n+a*n+b))*A[n] - F(n*(c*n*n+d))*A[n-1])
             / F((n+1)**3))
y0 = A[:N+1]

# 2. the log-solution tail g:  L(g) = -sum_j t^j P_j'(theta) y0
#    with P_0 = th^3, P_1 = -(2th+1)(a th^2 + a th + b),
#         P_2 = (th+1)(c(th+1)^2 + d)          [solve coefficientwise]
g = gseries(y0)

# 3. the nome and the mirror map
qt = smul([F(0),F(1)]+[F(0)]*(N-1), sexp(smul(g, sinv(y0))))  # q(t)
tq = srevert(qt)                                              # t(q)
Fq = compose(y0, tq)                                          # F(q)

# 4. sigma = theta_q log t,  P3 = 1 - 2a t + c t^2
T   = [tq[i+1] for i in range(N)] + [F(0)]                    # t/q
sig = smul([F(i)*T[i] for i in range(N+1)], sinv(T)); sig[0] += F(1)
t2  = smul(tq, tq)
P3  = [F(-2*a)*tq[i] + F(c)*t2[i] for i in range(N+1)];  P3[0] = F(1)

# 5. the companion
Psi   = smul(smul(tq, smul(sig, smul(sig, sig))),
             smul(sinv(P3), sinv(Fq)))
Theta = [F(0)] + [Psi[m]/F(m)**3 for m in range(1, N+1)]
Bn    = compose(smul(Fq, Theta), qt)     # Bn[n] = B(n);  b_n = 6*B(n)
```

Step 2 is the only one needing a word: writing $L = \sum_j t^j P_j(\theta_t)$, one solves (2.3) for g by reading off t^N, using that $P_0(N) = N^3 \neq 0$ for $N \geq 1$:

$$g_N = \frac{1}{N^3}\Big(R_N - \sum_{j\geq 1} P_j(N-j)\, g_{N-j}\Big), \qquad R = -\sum_j t^j P_j'(\theta_t) y_0.$$

Substituting steps 1–5 for any other Apéry-like (a, b, c, d) is the whole generalisation.

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[8] *Modular anchors for the deformation-unreachable sporadic families*, working paper of the present programme (`papers_out/modular_anchors`; probe notes `work/MODULAR_COACTION_PROBE.md`, scripts `work/z5eps/eps48_modular_nome.py`, `eps49_zeta_companion.py`).

[9] *Modularity and Yukawa integrality of the $\zeta(5)$ five-block*, working notes of the present programme (`work/Z5_MODULARITY_PROBE.md`).

[10] *The variational anchor*, working paper of the present programme (`papers_out/variational_anchor`).

[11] *The fifteen sporadic Apéry-like pairs: second solutions, the χ-twisted Lucas law, and the closed forms*, working paper of the present programme (`papers_out/sporadics`).